General Mathematics — Exercise Sheet 42

Gauss and Stokes

Variant: Statements

Exercise 1 (A tetrahedron's oblique face). Let $K = \{x, y, z \geq 0 : x + y + z \leq 1\}$ and $\boldsymbol{F} = (x, y, z)$. Compute total outward flux using the general divergence theorem [S], then verify it directly on all four faces [P]. Use Lecture 39's volume calculation and Lecture 41's graph vector area factor.

Exercise 2 (A cylinder with three boundary pieces). Let $R, H > 0$, $K = \{x^2 + y^2 \leq R^2, 0 \leq z \leq H\}$ and $\boldsymbol{F} = (xz, yz, z^2)$. Compute outward flux by Gauss [S] and directly on the top, bottom and curved side [P]. Give the normal choice on each piece. Use Lectures 39, 41 and 42.

Exercise 3 (A point singularity away from the centre). Let $R > 0$, $a \in \mathbb{R}^3$, $\kappa \in \mathbb{R}$ and $\boldsymbol{F}(x) = \kappa(x - a)/\|x - a\|^3$ for $x \neq a$. Use Gauss with the necessary puncture to find outward flux through the sphere $\|x\| = R$ when $\|a\| < R$ or $\|a\| > R$. Explain what fails in this theorem route when $\|a\| = R$ and $\kappa \neq 0$. If $\kappa = 0$, use the smooth zero extension and include all positions of a. Use Lecture 42 and a translation of its small-sphere calculation.

Exercise 4 (Stokes on a tilted graph). Let S be $z = 1 - x - y$ over $x^2 + y^2 \leq 1$, oriented upward, and $\boldsymbol{F} = (-yz, xz, 0)$. Compute curl flux directly and boundary circulation directly. Check their orientation agreement with Stokes [C]. Use Lectures 38, 41 and 42.

Exercise 5 (An obstruction to a vector potential). Let $\boldsymbol{H}(x) = x/\|x\|^3$ for $x \neq 0$. Show there is no C^1 vector field $\boldsymbol{A}$ on an open neighbourhood of the entire unit sphere for which $\operatorname{curl} \boldsymbol{A} = \boldsymbol{H}$ there. Use the cited Stokes theorem [C] on the whole closed sphere and the direct inverse-square flux calculation in Lecture 42. Retain the cited dependence.

Exercise 6 (Oriented solid angle). For an oriented compact surface S avoiding the origin, define its *oriented solid angle* by

$$\Omega(S) = \iint_S \frac{x \cdot \boldsymbol{n}}{\|x\|^3}\, \mathrm{d}S.$$

For $h, R > 0$, compute this quantity for the disc $z = h$, $x^2 + y^2 \le R^2$, with upward normal. Compare it directly with the outward spherical cap $0 \le \theta \le \theta_0$, where $\theta_0 = \arctan(R/h)$, on any sphere centred at the origin. Find the disc's limits as $R \downarrow 0$ and $R \to \infty$ with fixed h. Use Lectures 38–39, 41 and the definition, without applying Gauss to a cone containing its singular vertex.